# A4 Problems: Geometry and measurement

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**Draw a sketch for every problem.** When you find an angle, give the reason (for example "vertically opposite", "Z-angles", "angle sum of a triangle"). A calculator is allowed for problems with π or square roots; give exact answers (like $49\pi$ or $\sqrt{74}$) where asked, and otherwise round to one decimal.

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## Practice

**1.** Lines $\ell$ and $m$ are parallel, with $\ell$ above $m$. A line $t$ crosses $\ell$ at $P$ and $m$ at $Q$, going up to the right. At $P$, the angle between the part of $\ell$ to the right of $P$ and the part of $t$ above $P$ is $65°$. Find, with a reason each time:

(a) at $P$, the angle between $\ell$ to the left and $t$ upwards;
(b) at $P$, the angle between $\ell$ to the left and $t$ downwards;
(c) at $Q$, the angle between $m$ to the right and $t$ upwards;
(d) at $Q$, the angle between $m$ to the left and $t$ upwards.

**2.** Find the unknown angles.

(a) The angles of a triangle are $x$, $2x$ and $3x$.
(b) A triangle has angles $47°$ and $68°$. Find the third.
(c) A right-angled triangle has an angle of $34°$. Find the other two.
(d) In triangle $ABC$, the exterior angle at $C$ is $130°$ and $\angle A = 55°$. Find $\angle B$ and $\angle C$.

**3.** (a) What is the angle sum of a 10-gon (decagon)?

(b) What is each interior angle of a regular 12-gon?
(c) What is each exterior angle of a regular pentagon? And each interior angle?

**4.** Find the area.

(a) A triangle with base 10 cm and height 6 cm.
(b) A parallelogram with sides 8 cm and 6 cm; the height on the side of 8 cm is 5 cm.
(c) A trapezium with parallel sides 6 cm and 10 cm, and height 4 cm.

**5.** A circle has diameter 14 cm. Find its circumference and area, exactly (in terms of π) and rounded to one decimal.

**6.** (a) A cuboid (*balk*) measures 4 cm by 5 cm by 6 cm. Find its volume and its surface area.

(b) A cylinder has radius 3 cm and height 10 cm. Find its volume exactly and in litres (rounded to 2 decimals), and its total surface area exactly.

**7.** Triangle $ABC$ has $A(-2, 3)$, $B(4, 1)$, $C(1, -2)$. Give the coordinates of the image after:

(a) reflection in the $y$-axis;
(b) reflection in the $x$-axis;
(c) rotation through $90°$ anticlockwise about the origin;
(d) translation by $(3, -1)$.

**8.** For each shape give the number of lines of symmetry and the order of rotational symmetry (order 1 means no rotational symmetry). Which have point symmetry? square · rectangle (not a square) · regular pentagon · parallelogram (not a rectangle or rhombus) · isosceles trapezium · the letters H, N, A

**9.** (a) A right-angled triangle has legs 9 and 12. Find the hypotenuse.

(b) A right-angled triangle has hypotenuse 17 and one leg 8. Find the other leg.
(c) Is a triangle with sides 7, 24, 25 right-angled?

(d) Is a triangle with sides 5, 6, 8 right-angled?

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## School-exam level

**10.** (a) An isosceles triangle has a top angle of $36°$. Find the base angles.

(b) An isosceles triangle has a base angle of $50°$. Find the top angle.
(c) An isosceles triangle has an angle of $100°$. Find the other two angles, and explain why there is only one possibility.

**11.** A floor plan has the shape of a 10 m by 8 m rectangle with a 4 m by 3 m rectangle cut out of one corner (an L-shape).

(a) Find the area of the floor.
(b) Find the perimeter of the floor. Explain why it's the same as the perimeter of the full rectangle.

**12.** A ladder 5 m long leans against a vertical wall. Its foot is 1.4 m from the wall.

(a) How high up the wall does it reach?
(b) How far must the foot be moved away from the wall for the top to be 4 m high?

**13.** $A(1, 2)$, $B(7, 10)$ and $C(7, 2)$.

(a) Find the length of $AB$ and the midpoint of $AB$.
(b) Show that triangle $ABC$ is right-angled, and find its area.

**14.** A bicycle wheel has diameter 70 cm. How many full turns does it make on a ride of 1 km?

**15.** A swimming pool is 25 m long, 10 m wide and 2 m deep.

(a) How many litres of water does it hold when full?
(b) It is filled at 20 litres per second. How long does that take? Give hours and minutes.

**16.** Use only a compass and a straightedge (and a ruler to measure given lengths).

(a) Construct a triangle with sides 6 cm, 5 cm and 4 cm. Describe your steps.
(b) Explain why you can't construct a triangle with sides 2 cm, 3 cm and 6 cm.
(c) Draw a segment $AB$ of 7 cm and construct its perpendicular bisector. Mark a point $P$ on it and check that $PA = PB$. Explain why this is always so.
(d) Construct angles of $30°$ and $45°$.

**17.** (a) How many diagonals does a hexagon have? Draw them.

(b) Explain why an $n$-gon has $\dfrac{n(n-3)}{2}$ diagonals.
(c) How many diagonals does a 10-gon have?
(d) Which polygon has exactly 20 diagonals?

**18.** (a) Each interior angle of a regular polygon is $156°$. How many sides does it have?

(b) Is there a regular polygon with interior angles of $100°$? Explain.

**19.** (a) Find the area of the ring between two circles with the same centre and radii 5 cm and 3 cm.

(b) A circular pond has radius 4 m. A path 1 m wide runs around it. Find the area of the path, exactly and rounded to one decimal.

**20.** (a) What is the angle between the hands of a clock at 3:40? (Hint: the hour hand also moves: it turns $0.5°$ per minute.)

(b) At what time between 3:00 and 4:00 are the hands exactly on top of each other? Give the answer to the nearest second.

## Challenge

**21. Thales' theorem.** $AB$ is a diameter of a circle with centre $M$, and $C$ is any other point on the circle. Prove that $\angle ACB = 90°$. (Hint: draw $MC$. What kind of triangles are $AMC$ and $BMC$?)

**22.** (a) Use the formula for the interior angle sum to prove that the exterior angles of a convex $n$-gon add up to $360°$.

(b) Every interior angle of a regular polygon is $140°$. How many sides does it have? Use exterior angles.

**23.** Copies of one regular polygon are to cover the plane without gaps or overlaps, meeting corner to corner. Prove that this is only possible for equilateral triangles, squares and regular hexagons.

**24.** A $3 \times 3 \times 3$ cube is painted red on the outside and then cut into 27 small cubes.

(a) How many small cubes have 3, 2, 1 and 0 red faces?
(b) Answer the same for an $n \times n \times n$ cube (with $n \geq 2$). Check that your four answers add up to $n^3$.

**25.** In 1876 James Garfield, later a president of the United States, found this proof of Pythagoras' theorem. Take a right-angled triangle with legs $a$, $b$ and hypotenuse $c$. Place two copies and a third triangle to form a trapezium: put the first triangle with leg $a$ vertical and leg $b$ horizontal, then the second triangle on top of it with leg $b$ vertical and leg $a$ horizontal, so that the vertical legs together form a straight side of length $a + b$, and join the two free corners.

(a) Show that the third triangle (formed by the two hypotenuses and the joining line) has a right angle between the two hypotenuses.
(b) Calculate the area of the trapezium in two ways and deduce $a^2 + b^2 = c^2$.

**26.** An ant sits at one corner of a closed box measuring 3 cm by 4 cm by 5 cm. It wants to walk along the surface to the opposite corner (the one furthest away).

(a) How long is the shortest path? (Hint: unfold the box flat. There are several ways to unfold; compare them.)
(b) How long would the path be if the ant could fly straight through the box?

**27.** (a) Find the area of the triangle with vertices $A(1, 1)$, $B(7, 3)$, $C(3, 8)$ by drawing the smallest rectangle around it with sides parallel to the axes.

(b) Find the area of the triangle with vertices $(0, 0)$, $(6, 0)$ and $(2, 5)$, in two different ways.

**28.** A square has side 10. A circle is drawn inside it, touching all four sides. Inside the circle, a square is drawn with its four corners on the circle.

(a) Find the areas of the big square, the circle and the small square.
(b) What fraction of the big square does the circle cover? Does this depend on the size of the square?

**29. Archimedes' method.** Take a circle with radius $r$, so its circumference is $2\pi r$.

(a) A regular hexagon has its six corners on the circle. Explain why each side of the hexagon is $r$, and deduce that $\pi > 3$.
(b) A square is drawn around the circle, with each side touching the circle. Deduce that $\pi < 4$.
(c) Now draw a regular hexagon *around* the circle, each side touching the circle. Show that each side is $\dfrac{2r}{\sqrt{3}}$ and deduce that $\pi < 2\sqrt{3} \approx 3.46$.